

Duke University Phys 627/ECE 523 (Klco)
Exercise Set 2026

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I. WARM-UP AND PRELIMINARIES

A. tensor products

Nature appears to be constructed such that every quantum particle resides in its own linear vector space. Combining these linear vector spaces in order to describe many-particle quantum systems is achieved mathematically through the tensor product operation.

1. If a state is *separable* or *unentangled*, its state may be succinctly written in tensor product form, $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$. For a set of n two-dimensional Hilbert spaces, how many real numbers are needed to specify such an unentangled wavefunction? For an *entangled* state

that cannot be broken into such pieces, how many real numbers are needed to specify its wavefunction? How does this scaling change for d -level quantum degrees of freedom?

2. Declaring the basis in Dirac notation ($|000\rangle, \dots, |111\rangle$), write the vector representation of $|\psi\rangle^{\otimes 3}$ for $|\psi\rangle = \frac{|0\rangle + \sqrt{2}|1\rangle}{\sqrt{3}}$.
3. Algorithmically describe how to implement the tensor product operation.
 - Discuss the relationship between binary ordering of a Hilbert space basis and the tensor product operation. If two binary-ordered Hilbert spaces are expressed jointly, $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, what is the natural ordering of the resulting $(d_1 \times d_2)$ -dimensional Hilbert space?
 - Use this understanding to write (or generate) the matrix form of the unitary operators (extending the SWAP operator) that reorder the Hilbert spaces in the following ways:
 - $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3 \rightarrow \mathcal{H}_3 \otimes \mathcal{H}_2 \otimes \mathcal{H}_1$ with $d_{1,2,3} = 2$
 - $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3 \otimes \mathcal{H}_4 \rightarrow \mathcal{H}_2 \otimes \mathcal{H}_3 \otimes \mathcal{H}_1 \otimes \mathcal{H}_4$ with $d_{1,2,3,4} = 2$
 - $\mathcal{H}_1 \otimes \mathcal{H}_2 \rightarrow \mathcal{H}_2 \otimes \mathcal{H}_1$ with $d_{1,2} = 3$

Express these unitary operators in matrix form and as a list of $\prod_i d_i$ non-zero matrix elements (coefficients of outer product representation).

4. By hand, write the matrix representations of $X \otimes Y$, $\mathbb{I}_2 \otimes Y$, $Y \otimes \mathbb{I}_2$, and $Y \otimes \mathbb{I}_4$, where the Pauli matrices are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (1)$$

and $\mathbb{I}_k$ is the k -dimensional Identity matrix. Implement the tensor product operator in code to produce a picture of the matrix $Z \otimes X \otimes Y \otimes Z \otimes Y + X \otimes \mathbb{I}_2 \otimes Y \otimes Z \otimes Y$.

5. Due to the rapid growth in dimensionality of Hilbert spaces connected through the tensor product operation, structural computational design can have significant impact on computational resources. In some cases, the generation or manipulation of entanglement provides opportunities and guidance to improve such designs.

- Show that, for the application of a tensor product operator (e.g., non-entangling time evolution), $\mathbf{A} \otimes \mathbf{B}$ on a state vector, $\mathbf{v}$, one may employ the following identity

$$[\mathbf{A} \otimes \mathbf{B}] \mathbf{v} = \text{vec} [\mathbf{BVA}^T] \quad , \quad (2)$$

where $\mathbf{V}$ is a matrix form of the vector $\mathbf{v}$ that is reversed by the operation $\text{vec}[\cdot]$. (*May be helpful to begin with a separable $\mathbf{v}$ then employ the linearity of quantum mechanics*)

- For the case of $\dim[\mathbf{A}] = \dim[\mathbf{B}] = 10^3$ (i.e., $\mathbf{A}$ and $\mathbf{B}$ are $10^3 \times 10^3$ dimensional matrices), estimate the percent reduction in the number of floating point operations used to act the tensor-product operator on the state vector when employing this identity.

B. Gram-Schmidt orthogonalization

The Gram-Schmidt procedure produces an orthonormal basis, $\{|v_k\rangle\}$, in a d -dimensional vector space from an arbitrary spanning basis, $\{|w_k\rangle\}$. Upon choosing a vector in the basis as the initial seed, $|v_1\rangle = \frac{|w_1\rangle}{\sqrt{\langle w_1|w_1\rangle}}$, remaining vectors are sequentially orthogonalized from the collection as

$$|v'_k\rangle \equiv |w_k\rangle - \sum_{j=1}^{k-1} \langle v_j|w_k\rangle |v_j\rangle \quad |v_k\rangle = \frac{|v'_k\rangle}{\sqrt{\langle v'_k|v'_k\rangle}} \quad , \quad (3)$$

for $2 \leq k \leq d$.

1. Show that the Gram-Schmidt process produces an orthonormal basis for the vector space.
2. Determine (recommend teaching a computer how to calculate) the orthonormal basis identified through GS orthogonalization of the following $\{|w_k\rangle\}$ sets

- $\{|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle\}$
- $\{|\psi_4\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle\}$
- $\{|\psi_3\rangle, |\psi_4\rangle, |\psi_1\rangle, |\psi_2\rangle\}$
- $\{|\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_1\rangle\}$

with states

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -i \end{pmatrix} \quad |\psi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad |\psi_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \quad |\psi_4\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad . \quad (4)$$

C. Hermitian and unitary matrices

1. Define the properties of Hermitian and unitary matrices. Discuss which properties make Hermitian and unitary matrices logical candidates for representing physical properties of energy spectra (as Hamiltonian) and propagation (as time evolution operators), respectively.
2. Solve the Schrödinger equation for a time-independent Hamiltonian to determine the relation between the Hamiltonian and the time evolution operator.
3. Show that: for every Hermitian matrix, A , the exponential e^{iA} is unitary. (*The series expansion may be useful*)
4. Show that: every unitary matrix can be written as the imaginary exponential of a Hermitian matrix, e^{iH} , and can thus be considered as the time evolution of some Hamiltonian. (*Consider their spectral decompositions*)
5. **Ehrenfest's theorem** Show that the commutation of a time-independent operator with the Hamiltonian provides a necessary and sufficient condition for its expectation value to be constant in time.
6. Discuss when a unitary transformation of time evolution can be encoded as a unitary transformation of the associated Hamiltonian, $Ue^{iH}U^\dagger = e^{iUHU^\dagger}$.

7. Given the Schrödinger equation ($\hbar = 1$) with time-independent Hamiltonian,

$$i \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle \quad , \quad (5)$$

find the equation of motion (time derivative) for the time-evolution of the density matrix,

$$\rho(t) = \sum_{j=1}^n p_j |\psi_j(t)\rangle \langle \psi_j(t)| \quad (6)$$

Does the density matrix evolve like a Heisenberg picture operator?

D. Pauli matrices

Operators for spin- $\frac{1}{2}$ quantum degrees of freedom can be represented as 2×2 matrices.

1. Pauli matrices are matrix representations of Pauli operators with respect to the “ Z ”-basis, $|0\rangle$ and $|1\rangle$,

$$(\sigma_0, \vec{\sigma}) = (\sigma_0, \sigma_1, \sigma_2, \sigma_3) = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) \quad , \quad (7)$$

Express each of these Pauli matrices in outer product notation. How does each Pauli matrix transform the two orthonormal basis states, $|0\rangle$ and $|1\rangle$.

2. Using the Kronecker delta and Levi-Civita epsilon in index notation, determine succinct expressions for the commutators, $[\sigma_i, \sigma_j]$, the anticommutators, $\{\sigma_i, \sigma_j\}$, and the operator products, $\sigma_i \sigma_j$, for $i, j \in \{1, 2, 3\}$. Use this to determine the maximum number of Pauli operators that can be simultaneously represented as diagonal matrices.

3. Let $\vec{v}$ be any real three-dimensional unit vector and θ a real number. Show that

$$e^{i\theta \vec{v} \cdot \vec{\sigma}} = \cos(\theta) \mathbb{I}_2 + i \sin(\theta) \vec{v} \cdot \vec{\sigma} \quad , \quad (8)$$

where $\vec{v} \cdot \vec{\sigma} \equiv \sum_{j=1}^3 v_j \sigma_j$. Generalize this to show that for any function, $f(\cdot)$, that maps complex numbers to complex numbers,

$$f(\theta \vec{v} \cdot \vec{\sigma}) = \frac{f(\theta) + f(-\theta)}{2} \mathbb{I} + \frac{f(\theta) - f(-\theta)}{2} \vec{v} \cdot \vec{\sigma} \quad . \quad (9)$$

4. Pauli decompositions are often utilized in quantum computing to break computations into pieces that are manageable to implement on quantum devices. Convince yourself that a complete basis for representing generic spin- $\frac{1}{2}$ Hermitian operators is spanned by the Pauli matrices, i.e., all single-qubit operators can be expressed as a vector of Pauli coefficients. Utilizing the trace properties of the Pauli operators, devise a systematic way of determining the Pauli decomposition of an arbitrary single-qubit operator, H_2 . Extend the technique to arbitrary n -qubit operators, H_{2^n} . Implement your scheme to determine the multi-qubit Pauli decomposition of the four-qubit operators

$$\hat{\phi}^{(JLP)} = \frac{1}{15} \text{diag}(-15, -13, -11, -9, -7, -5, -3, -1, 1, 3, 5, 7, 9, 11, 13, 15) \quad (10)$$

and

$$\hat{\phi}^{(HO)} = \frac{a + a^\dagger}{\sqrt{2}} \quad , \quad (11)$$

where $\hat{\phi}^{(HO)}$ is defined by matrix elements $\hat{\phi}_{j,j+1} = \hat{\phi}_{j+1,j} = \sqrt{j+1} \forall j \in \{0, 14\}$. How does the number of terms in these two decompositions scale with the number of qubits spanned by the operator? For an arbitrary operator, what is the worst-case scaling of the number of terms in its Pauli decomposition?

E. Connecting finite and infinite dimensions

The quantum harmonic oscillator Hamiltonian, $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2}\hat{x}^2$ may be expressed in terms of dimensionless variables, $\bar{p} = \frac{\hat{p}}{\sqrt{\hbar m \omega}}$ and $\bar{x} = \sqrt{\frac{m\omega}{\hbar}}\hat{x}$, as

$$\hat{H} = \frac{\hbar\omega}{2} (\bar{p}^2 + \bar{x}^2) \quad . \quad (12)$$

In this exercise, you will calculate (confirm) the harmonic oscillator ground state energy in four complementary ways. The first three are expected to be simple review.

1. Ladder operators:

Considering the operators

$$\hat{a} = \frac{\bar{x} + i\bar{p}}{\sqrt{2}} \quad \hat{a}^\dagger = \frac{\bar{x} - i\bar{p}}{\sqrt{2}} \quad , \quad (13)$$

propagate the position-momentum canonical commutation relation to determine $[\hat{a}, \hat{a}^\dagger]$ and $[\bar{x}, \bar{p}]$. By writing the Hamiltonian in terms of these ladder operators, calculate the ground state energy expectation value as $\langle 0 | \hat{H} | 0 \rangle$ where $|0\rangle$ is the ground state.

2. Finite truncation in number basis (qudit):

In the basis formed by eigenstates of the number operator, $\hat{N} = \hat{a}^\dagger \hat{a}$, write the matrix representation (or a pattern-extendable low-energy subspace) for the operators $\hat{N}$, $\hat{a}$, and $\hat{a}^\dagger$. In the ordering of your basis, what is the vector form of $|0\rangle$? Truncating these matrices at increasing dimensionalities Λ to systematically include higher energy states, show the convergence of $\langle 0 | \hat{H} | 0 \rangle$ as a function of Λ . Discuss the form of the commutator in these finite truncations, and connect your observation to the Stone-von Neumann theorem through the trace of the canonical commutation relation.

3. Continuous wavefunction:

You saw above that the ground state of the harmonic oscillator is a computational basis state when expressed in the number operator basis. Fill in the steps to Sakurai textbook Eq. (2.3.30) to show that the ground state expressed in the continuous position basis has a Gaussian wavefunction, $\psi(x) = \langle x | 0 \rangle \propto e^{-\frac{\bar{x}^2}{2}}$. Again, calculate the ground state energy expectation value, this time expressed as an integral in position space.

4. Basis of position eigenstates:

Consider a basis of position eigenstates organized into a grid characterized by symmetric upper/lower bounds x_{max} and lattice spacing δ_x . First, show that an approximate representation of the squared momentum operator in this basis is $\bar{p}^2 = \left(-i\frac{\partial}{\partial \bar{x}}\right)^2 = \frac{A}{\delta_x^2}$, where A is a

Toeplitz matrix with $a_0 = 2$ and $a_1 = a_{-1} = -1$, e.g., via matrix multiplication of the forward and backward leading-order finite difference approximations of the first derivative. The matrix representation of $\hat{x}$ in this basis is diagonal with matrix elements spanning $\pm x_{max}$ with a spacing of δ_x . With a relationship of $\delta_x = \frac{2x_{max}}{n_{states}-1}$, show that the ground state energy of this Hamiltonian approaches its continuum value as the number of states increases. Plot the associated ground state wavefunctions, are they as expected? have you chosen an x_{max} that captures its spatial extent?

5. Consider implications of these results for qubit implementations with respect to the Pauli decompositions performed at the end of Exercise [1D](#). If the quantum simulation cost scales with the number and size of terms in the Pauli decomposition of the Hamiltonian, which basis might you choose for digital representation of a harmonic oscillator-type Hilbert space on a quantum computer?

II. ALGORITHMIC COMPONENTS

A. Hadamard operator as transform

The Hadamard gate has a matrix form of $\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| - |1\rangle\langle 1|)$.

1. Verify that the Hadamard is Unitary and is its own Hermitian conjugate.
2. Calculate the Hadamard transformed Pauli operators: HXH , HYH , and HZH . Do your results make sense in light of the Hadamard's role as the $x \leftrightarrow z$ basis transformation, $H = |0\rangle\langle +| + |1\rangle\langle -|$?
3. Show that the single-qubit Hadamard operator over n -qubits, $H^{\otimes n}$, can be written as

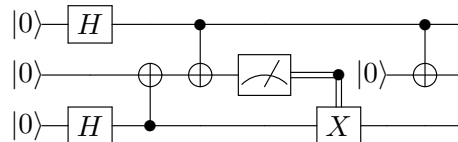
$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} |x\rangle\langle y| \quad , \quad (14)$$

where $|x\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle$ and $x = x_1x_2 \cdots x_n$ is a binary representation of x , and likewise for y , leading $x \cdot y$ to be the bit-wise inner product.

4. Write and describe the state $H^{\otimes n}|0\rangle$, where $|0\rangle = |0\rangle_1|0\rangle_2 \cdots |0\rangle_n$.

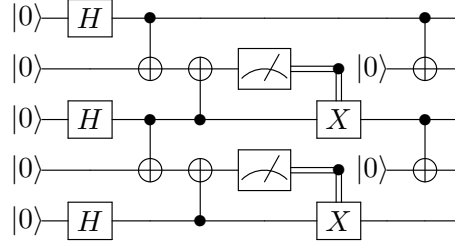
B. GHZ state preparation

1. With one Hadamard operator and $n - 1$ CNOTs, write a Unitary circuit that is capable of preparing the n -qubit GHZ state. What is the depth of your circuit? On average, what fraction of time is each qubit idle?
2. Analyze step-by-step the functionality of the following circuit.



Is the final state dependent on the outcome of the measurement?

3. Analyze step-by-step the functionality of the following extension of the circuit above.



If this circuit is extended to generate an n -qubit GHZ state, how many CNOTs are used? Describe the strategy of this circuit in the language of measuring and manipulating (CS)parity.

4. Refer to [PRX Quantum 5, 030339](#) to contextualize the significance of this observation. Discuss the value of mid-circuit measurement and feedforward beyond quantum error correction.

C. Deutsch-Jozsa algorithm

As discussed interferometrically in the first lecture, the Deutsch Jozsa algorithm may be represented with the following quantum circuit

$$\begin{array}{c} |0\rangle^{\otimes n} \xrightarrow{H^{\otimes n}} \\ |-\rangle \end{array} \xrightarrow{U_f} \xrightarrow{H^{\otimes n}} \text{Measurement} \quad , \quad (15)$$

where $U_f|\mathbf{x}\rangle|y\rangle = |\mathbf{x}\rangle|y \oplus f(\mathbf{x})\rangle$.

1. Show that $|y\rangle = |-\rangle$ is an eigenstate of the controlled operator, U_f , with eigenvalue $(-1)^{f(\mathbf{x})}$.
2. Utilizing the multi-qubit local Hadamard transform, analyze the above circuit to find the final state prior to measurement. Describe the logic applied to the measurement results that allows a single implementation of this circuit (i.e., not ensemble measurement) to determine whether the function is constant, $f(\mathbf{x}) = c \forall \mathbf{x} \in \{0, 1\}^{\otimes n}$ and $c \in \{0, 1\}$, or the function is balanced with an equal number of 0 and 1 evaluations across the 2^n inputs. Remark upon the state remaining on the last qubit after measurement.
3. For each of the four possible single-qubit constant/balanced binary functions,

$$f(0) = f(1) = 0 \quad (16)$$

$$f(0) = f(1) = 1 \quad (17)$$

$$f(0) = f(1) \oplus 1 = 0 \quad (18)$$

$$f(0) = f(1) \oplus 1 = 1 \quad , \quad (19)$$

write a circuit representation of the two-qubit gate U_f .

4. For each of the eight possible two-qubit constant/balanced binary functions,

$$(f(0) \ f(1) \ f(2) \ f(3)) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix} \quad , \quad (20)$$

write a circuit representation of the three-qubit gate U_f .

5. For an n -bit application, $f : \{0,1\}^n \rightarrow \{0,1\}$, calculate the total number of possible constant/balanced binary functions. How does this number scale asymptotically with n . Describe one circuit compilation strategy implementing U_f for arbitrary n . (*Consider which functions are simple to compile vs those that require additional circuit logic.*) In the worst case, how does the depth of your circuit scale with n ?

D. quantum Fourier transform identity

In the interferometry perspective of quantum computation, we interpreted the role of the Hadamard transform as delocalizing the wavefunction in Hilbert space. Generalizing this perspective to an operation that transfers delta-functions to constant-functions between conjugate Hilbert spaces (e.g., position-momentum) leads to the Quantum Fourier Transform. The following product-structured identity is crucial for creating and understanding the Quantum Fourier Transform circuit,

$$QFT|\mathbf{x}\rangle = \frac{1}{\sqrt{2^n}} \sum_{\mathbf{k}=0}^{2^n-1} e^{\frac{2\pi i \mathbf{x} \cdot \mathbf{k}}{2^n}} |\mathbf{k}\rangle \quad (21)$$

$$= \frac{|0\rangle + e^{2\pi i(0.x_n)}|1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + e^{2\pi i(0.x_{n-1}x_n)}|1\rangle}{\sqrt{2}} \otimes \dots \otimes \frac{|0\rangle + e^{2\pi i(0.x_1x_2\dots x_n)}|1\rangle}{\sqrt{2}} \quad (22)$$

In the binary notation used here, integers are notated as $\mathbf{x} = x_1x_2\dots x_n = x_12^{n-1} + \dots + x_n2^0$ and fractions are notated as $(0.x_{n+1}x_{n+2}\dots x_{n+m}) = \frac{x_{n+1}}{2} + \frac{x_{n+2}}{2^2} + \dots + \frac{x_{n+m}}{2^m}$.

1. Show that this identity is true.
2. Use this form to motivate explicit QFT circuit implementations for 1, 2 and 3 qubits.

Extra Using the QFT in its role of transforming operators between momentum and position space, show how the local finite-difference momentum operator of [IE.4](#) can be replaced by a proper nonlocal lattice momentum operator to achieve more rapid convergence of the ground state energy.

III. QUANTUM INFORMATION PHENOMENA

A. uncertainty principle and enhanced sensing

Consider two Hermitian ($\hat{A} = \hat{A}^\dagger = (\hat{A}^*)^T$) operators, $\hat{A}$ and $\hat{B}$, whose commutator is non-vanishing.

1. Show that there is an inherent uncertainty in the associated expectation values

$$\Delta(\hat{A})\Delta(\hat{B}) \geq \frac{|\langle [\hat{A}, \hat{B}] \rangle|}{2} \quad (23)$$

where $\Delta(\hat{A}) = \sqrt{\langle (\hat{A} - \bar{\hat{A}})^2 \rangle}$ is the standard deviation from the average, $\bar{\hat{A}} = \langle \hat{A} \rangle$. For position and momentum, $[\hat{x}, \hat{p}] = i\hbar$, this reproduces the familiar $\Delta(\hat{x})\Delta(\hat{p}) \geq \hbar/2$. **Hint** you may find useful the Cauchy-Schwartz inequality for states in an inner product vector space: $\langle u|u\rangle\langle v|v\rangle \geq |\langle u|v\rangle|^2$.

2. Discuss how this relationship limits the minimum uncertainty achievable when measuring two conjugate variables. Further, describe the process through which fluctuations can be re-distributed to reduce uncertainty in one variable for improved sensitivity in quantum sensing (see Phys. Rev. A 46, R6797(R) *Spin squeezing and reduced quantum noise in spectroscopy*).

B. no-cloning theorem

The no-cloning theorem states that it is impossible to design a quantum circuit that perfectly copies an unknown quantum state, i.e., there exists no U_{xerox} such that $U_{\text{xerox}}|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for arbitrary $|\psi\rangle$. More broadly, there exists no superoperator, $\mathcal{E}$, such that $\mathcal{E}(|\psi\rangle\langle\psi| \otimes |x\rangle\langle x|) = |\psi\rangle\langle\psi| \otimes |\psi\rangle\langle\psi|$ that produces additional copies of an arbitrary state $|\psi\rangle$.

1. Find a specific U_{xerox} that copies Z -axis basis states, $U_{\text{xerox}}|x\rangle|0\rangle = |x\rangle|x\rangle$ for $x \in \{0, 1\}$. What does this operator create when applied to more general superposition initial states?
2. By exploring the repercussions of the existence of such a U_{xerox} , devise a proof-by-contradiction for the no-cloning theorem.
3. Discuss physical implications of the no-cloning theorem, e.g., on state tomography, the uncertainty principle, quantum error correction, security in quantum cryptography, etc.
4. If a U_{xerox} did exist, utilize it to devise a super-luminal communications scheme.

N&C Exercise 1.2 Explain how a device that could perfectly distinguish between two non-orthogonal states could be used to create U_{xerox} to clone them. Or alternatively, show how U_{xerox} could be used to distinguish non-orthogonal states.

5. Choose an approximate quantum cloning machine (either one available “on the market” or create your own) and demonstrate its fidelity for the six eigenvectors of X , Y , and Z Pauli operators.

C. distinguishability of quantum states

1. When provided an ensemble of identical quantum states, numerous measurements in multiple bases may perform *quantum state tomography* to determine the wavefunction characterizing the ensemble. With access to only a single quantum measurement in any basis, show that two non-orthogonal pure states cannot be perfectly distinguished.
2. Your experimentalist friend devises a game in which one instance of a quantum state is randomly generated that is promised to be prepared in either $|\psi_1\rangle$ or $|\psi_2\rangle$. The two possible wavefunctions are provided and generated with an even 50/50 probability. You are allowed to projectively measure this instance in a basis of your choice prior to guessing the state prepared.

- Consider the two possible states of

$$|\psi_1\rangle = |0\rangle \quad |\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad . \quad (24)$$

If you chose to measure in the basis $\{|\psi_1\rangle, |\psi_1^\perp\rangle\}$, what would be your maximum probability of success? By rotating the measurement basis in the plane established by $\{|\psi_1\rangle, |\psi_2\rangle\}$, determine the maximum probability of success that can be achieved.

- Provide a geometric interpretation for the orientation of this optimal basis. Use this geometric reasoning to show that the maximum probability of success to distinguish an arbitrary pair of states will be $p_{\max \text{ success}} = \frac{1 + \sin(\arccos |\langle \psi_1 | \psi_2 \rangle|)}{2}$.
 - How is this optimal basis related to the difference of the density operators $D = \rho(|\psi_1\rangle) - \rho(|\psi_2\rangle)$? (*Hint:* see Holevo-Helstrom theorem)
 - Discuss how the odds of successful identification improve if the game provides an ensemble of two instances of the state. Should the same measurement basis be used for both measurements? If not, how would you choose the second measurement basis?
3. If possible, discuss experimentally how to distinguish between the following two mixed state ensembles

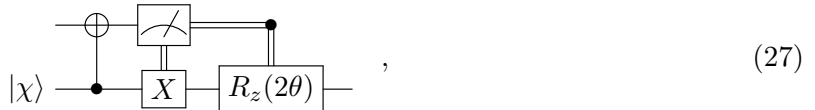
$$\rho_1 = \frac{1}{2}|\Phi^+\rangle\langle\Phi^+| + \frac{1}{2}|\Phi^-\rangle\langle\Phi^-| \quad (25)$$

$$\rho_2 = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11| \quad , \quad (26)$$

where $|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}$ are two entangled Bell states. If not possible, explain why.

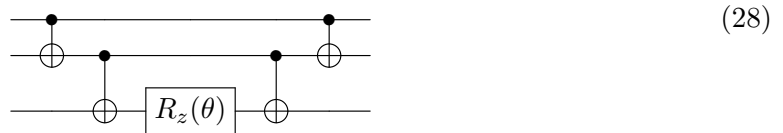
D. quantum gate teleportation and compilation

1. Analyze the function of the following circuit



where $|\chi\rangle = R_z(\theta)|+\rangle$. Taking $\theta = \pi/4$, discuss the importance of this circuit in the context of $\{H, T\}$ being universal for single-qubit unitaries and the fault tolerance available to the Clifford subgroup.

2. When quantum gate applications are probabilistic or imperfect, it can be advisable to “store” a successful application of a gate on an external qubit resource for future seeding of a gate teleportation circuit. For CNOT gates, this strategy can trade the task of implementing the two-qubit gate for the task of creating and verifying an entangled state. Fill in the details of the discussion in Section 5.3 of KLM, and complete associated exercise 5.3.1 establishing the necessary classical logic for implementing a CNOT via a four-qubit entangled state and single-qubit operators classically controlled by measurements in the Bell basis.
3. Describe the relationship between the Pauli rotation $e^{-iZ^{\otimes n}t}$ and the parity (number of ones mod 2) of the quantum state in the computational basis, i.e., basis in which Z is diagonal. From this perspective, analyze the following circuit



Describe the modifications, e.g., external basis changes, that produce circuits capable of implementing any Pauli rotation $e^{-i\hat{P}_v t}$.

IV. TOWARD QUANTUM SIMULATION

A. measuring purity

Measuring the purity of a quantum state $\text{Tr} [\rho^2]$ may be a valuable indicator of experimental quality or the presence of subsystem entanglement. Because the quantity is quadratic in the density matrix, it cannot be connected to any single expectation value.

- Tensor product Pauli operators, $\hat{P}_{\mathbf{v}} = \sigma_{v_1} \otimes \sigma_{v_2} \otimes \cdots \otimes \sigma_{v_n}$ with $\mathbf{v} = \{\mathbb{I}_2, \sigma_x, \sigma_y, \sigma_z\}^n$, provide a complete basis for decomposing density matrices. By writing the structure of an arbitrary density matrix with respect to expectation values of operators in this basis, report how many distinct measurement ensembles would be needed to learn the density matrix (tomography) as a function of n , the number of qubits in the state.
- There are many strategies for calculating the purity without performing full state tomography. For two copies of a single-qubit state, determine and physically describe the relationship between the expectation value $\langle \Psi^- | \rho \otimes \rho | \Psi^- \rangle$ and the purity of the state.
- For an arbitrary number of qubits, motivate the relationship $\text{Tr} [\rho^2] = \text{Tr} [(\rho \otimes \rho) \text{SWAP}]$. Using a basic interferometric quantum circuit structure (e.g., Hadamard test) for the non-destructive measurement of expectation values, show how the purity information may be calculated through the statistics of a single auxiliary qubit.

B. projecting spatial parity

Under the interchange of spatial coordinates, or equivalently the swapping of quantum states between Hilbert spaces, even-parity states are unchanged while odd-parity states accrue an overall minus sign. In the case of two spin-1/2 particles, these categories are also called “triplet” and “singlet” states, referring to the dimensionality of each within the four-dimensional Hilbert space. For such a two-qubit system, the positive parity (triplet) states are

$$|\psi_{t,1}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad |\psi_{t,2}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad |\psi_{t,3}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \quad , \quad (29)$$

while the negative parity (singlet) state is

$$|\psi_s\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \quad . \quad (30)$$

You will recognize these as components of the Bell basis.

- Show that under interchange of the two Hilbert spaces, the states have the symmetry properties as advertised.

In class, we discuss the non-destructive partial projective measurement of bit-wise parity by introducing an auxiliary qubit. The bit-wise parity was “computed” onto the auxiliary qubit via U_p , entangling the system and auxiliary Hilbert spaces. Subsequent measurement of the auxiliary qubit in state $|0\rangle/|1\rangle$ was shown to project the system into the 0/1 bit-wise parity subspace.

- What are the projectors of spatial parity, $P_{\pm}$?

- Devise a new quantum circuit, U_p , that computes the value of spatial parity onto an auxiliary qubit. Show that the effective system-space operators are equal to the parity projectors, $\hat{E}_0 = \langle 0|U_p|0\rangle = P_+$ and $\hat{E}_1 = \langle 1|U_p|0\rangle = P_-$, where the kets, as usual, reside in the auxiliary Hilbert space yielding an operator in the system Hilbert space.

Hint: Inspired by the interferometric circuit structure discussed on Day 1, consider the phase kickback available when conditionally applying the SWAP operator to the system, where

$$\text{SWAP} = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11| = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (31)$$

Can you coax this phase into producing a bit flip in the same way we were able to conclusively determine the difference between $\phi = 0, \pi$ with a single measurement at the two final detectors (e.g., Deutsch's algorithm)?

C. time evolution and error mitigation

1. **Quantum Zeno effect** Consider a two-dimensional quantum degree of freedom in the state $|S_z^+\rangle = |0\rangle$ that is subject to external fields whose effects can be modeled at leading order as a Z -basis-flip Hamiltonian, $\hat{H} = \frac{X+Y}{\sqrt{2}}$, with unit coupling constant $U_{ext} = e^{-i\frac{X+Y}{\sqrt{2}}t}$. If you aimed to retain the $|0\rangle$ quantum state through repeated measurement in the Z -basis, how many (non-destructive) measurements would you have to perform in order to produce a probability of at least 99% that the state has remained in the vicinity of the $|0\rangle$ state throughout one unit of time evolution?
2. **Symmetry Identification** Consider the slightly more interesting scenario of two such quantum degrees of freedom beginning in the Bell state $\frac{|00\rangle + |11\rangle}{\sqrt{2}}$ and subject to a unitary time evolution of the form $U_{\text{noiseless}} = e^{-i(X \otimes Y)t}$.
 - (a) Projected into the Bell basis, plot the evolution of probabilities for times up to $t = 2$.
 - (b) Identify an operator (e.g., parity and/or built from Bell basis projectors) whose expectation values distinguish the populated states from those that remain unpopulated.
 - (c) Show that this operator commutes with the Hamiltonian, and thus presents a static symmetry under time evolution.
3. **Coherent Errors: Zeno and Basis Randomization** Consider the presence of single-qubit errors such that the time evolution becomes $U_{\text{noisy}} = e^{-i\left(X \otimes Y + \mathbb{I} \otimes \frac{X+Y}{\sqrt{2}} + \frac{X+Y}{\sqrt{2}} \otimes \mathbb{I}\right)t}$.
 - (a) Plot the new evolution of probabilities projected into the Bell basis. For how long do the probabilities remain within $\pm 1\%$ of their noiseless values?
 - (b) Consider the procedure of periodically projectively measuring (non-destructively) the operator whose expectation values identify the positive-parity subspace, and post-selecting upon the results.
 - Describe experimentally how such measurements could be performed, either abstractly or in quantum system of your choice.

- Show that, at the cost of measurement statistics, this procedure can improve the fidelity of the noisy simulation. How many such measurements would be required to maintain probabilities within $\pm 1\%$ of their noiseless values to a simulation time of $t = 1.5$ with a loss of no more than 20% measurement statistics?
- (c) Consider an alternate error mitigation technique, *twirling*, that introduces a dynamic randomization of Pauli bases throughout the time evolution to turn coherent noise into incoherent channels, in some cases destructively interfering coherent noise entirely.
- Viable basis transformations of the noiseless operators leave invariant the logical circuits of time evolution. Find all Pauli operators, within the set of 16 $P_j = \{\mathbb{I}_2, X, Y, Z\} \otimes \{\mathbb{I}_2, X, Y, Z\}$, that satisfy the criteria

$$U_{\text{noiseless}} = P^\dagger U_{\text{noiseless}} P \quad . \quad (32)$$

- Devise a stochastic strategy of these invariant basis changes that maintains, up to $t = 1.5$, the Bell basis measurement probabilities within $\pm 10\%$. How many layers of basis changes were used? Is the utilization of all basis transformations necessary to achieve this goal? (*Hint:* while probabilistic numerical simulation could be used, try utilizing density matrix methods to capture the stochastic features of the procedure)

V. NATURE OF MIXED STATES

A. quantum sabotage

In class, we discuss the close relationship between unitarily establishing entanglement with systems that become experimentally inaccessible and errors/noise in quantum computational endeavors. In this exercise, you will have a chance to explore this relationship in further detail.

1. We show that an amplitude damping channel, $\mathcal{E}_{AD}(\rho)$, models spontaneous decay from the $|1\rangle$ state to the $|0\rangle$ state. When measuring the environment in the Z -basis, this channel can be decomposed into a set of operators,

$$\rho' = \mathcal{E}_{AD}(\rho) = E_0 \rho E_0^\dagger + E_1 \rho E_1^\dagger \quad E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad E_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix} \quad . \quad (33)$$

This *channel* represents the non-unitary interaction seen from the perspective of the qubit system alone. Choose a few pure states on the surface of the Bloch sphere and plot their trajectories as a function of time (or decay time steps) when subject to this type of environment interaction.

2. The phase damping channel may arise from fluctuations in unitary controls or scattering events that do not shift the state of the computational Hilbert space. Show that in the latter scenario, the qubit observes a channel of the form

$$\rho' = \mathcal{E}_{PD}(\rho) = E_0 \rho E_0^\dagger + E_1 \rho E_1^\dagger \quad E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \quad E_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{p} \end{pmatrix} \quad . \quad (34)$$

Again, choose a few pure states on the surface of the Bloch sphere and plot their trajectories as a function of time (or decay time steps) when subject to this type of environment interaction.

3. When analyzing the teleportation process from the perspective of the receiver, we recognized the depolarizing channel,

$$\mathcal{E}_D(\rho) = p \frac{\mathbb{I}}{2} + (1-p)\rho \quad . \quad (35)$$

One last time, choose a few pure states on the surface of the Bloch sphere and plot their trajectories as a function of time (or decay time steps) when subject to this type of environment interaction. In terms of the vector of Pauli matrices, $\vec{\sigma}$, find the local operators, E_k that express the depolarizing channel in the form $\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger$.

4. In a desire to erase your quantum hard drives, but not destroy the drives themselves, you decide to bring in destructable auxiliary qubits to implement the completely depolarizing channel ($p = 1$). For a single-qubit drive, construct a circuit describing a unitary interaction, U , between your drive and the auxiliary qubits that will produce this depolarizing channel.

Hint: what effect would a CNOT have?

Hint: you may consider using more than one qubit in the auxiliary qubit “environment”?

5. Generalize your unitary circuit to implement the the completely depolarizing channel to a two qubit system, $\mathcal{E}(\rho) = \frac{\mathbb{I}}{4}$. How would your circuit generalize to an n -qubit system, $\mathcal{E}(\rho) = \frac{\mathbb{I}}{2^n}$? How many additional qubits do you anticipate using as a function of this memory capacity of your drive?

B. non-uniqueness of convex decompositions

1. Consider the following pure state

$$|\psi\rangle = \frac{1}{\sqrt{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle \quad . \quad (36)$$

All pure-state density matrices have single-term convex decompositions such that $\rho = |\psi\rangle\langle\psi|$. The associated measurement probability vector in the density matrix eigenbasis is thus $\{1, 0, 0, \dots, 0\}$.

Find an alternate decomposition of the density matrix that is composed of an orthonormal basis yielding a uniform measurement probability vector.

2. Through generalization of your methods, argue that quantum states, pure or mixed, of any qubit number have an associated orthonormal basis in which measurement outcomes are uniform, and thus minimally informative.
3. This property of measurement bases for pure states is the trivial limiting case of a more general relation. Consider density matrix ρ with vector of eigenvalues $\vec{\lambda}_\rho$: there exists an orthonormal basis $\{|v\rangle\}$ decomposing the density matrix as $\rho = \sum_k p_k |v_k\rangle\langle v_k|$, if and only if $\vec{p}$ is *majorized* by $\vec{\lambda}_\rho$, $\vec{p} \prec \vec{\lambda}_\rho$. Physically, vector ℓ is majorized by vector τ when ℓ can be created by τ by averaging over random permutations of τ 's components. Majorization thus provides a hierarchy of disorder or mixedness. The uniform distribution is at the base of this hierarchy, consistent with the observation that the uniform distribution can be majorized by any other probability distribution.

After elaborating upon the definition, derivation, and physical meaning of this mathematical statement, use this information to

- state the minimum number of pure state ensembles that could be used to prepare a mixed state.
- determine whether an orthonormal basis exists associated with the following probability measurement vectors for the mixed state $\rho = \frac{1}{4}\mathbb{I}_4 + \frac{1}{8}(X \otimes X - Y \otimes Y)$:
 - $\{\frac{1}{2}, 0, 0, \frac{1}{2}\}$
 - $\{\frac{1}{3}, \frac{1}{3}, 0, \frac{1}{3}\}$
 - $\{0, \frac{1}{6}, \frac{2}{3}, \frac{1}{6}\}$
 - $\{\frac{1}{16}, \frac{3}{8}, \frac{1}{2}, \frac{1}{16}\}$
 - $\{\frac{2}{8}, \frac{3}{8}, \frac{2}{8}, \frac{1}{8}\}$

In other words: physically, if you aimed to prepare ρ in the lab and, upon scrutiny of your success, found an ensemble measurement basis yielding each of the above probability distributions, would this raise concern for your state preparation fidelity?

For further discussion on the role and importance of majorization in quantum information, see *Phys. Rev. Lett.*, Vol 83 (2), pp 436–439 (1999) by Nielsen and *Phys. Rev. Lett.*, Vol. 86, 5184–7 (2001) by Nielsen and Kempe.

C. subsystem entanglement quantification

You wish to distribute highly entangled states between two parties, A and B . However, your entanglement resource generators will only create three-qubit entangled states, $|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}$ or $|W\rangle = \frac{|001\rangle + |010\rangle + |100\rangle}{\sqrt{2}}$.

As we discuss, the GHZ-type of entanglement is “fragile” in the sense that tracing/ignoring one of its qubits leaves the remaining two in a mixed state with density matrix identical to that of a completely classical mixture, $\frac{|00\rangle\langle 00| + |11\rangle\langle 11|}{2}$. However, the $|W\rangle$ state, upon tracing of a qubit retains entanglement. Thus, if the third qubit is ignored, utilization of the $|W\rangle$ state generator would be advised.

In the following, we will explore whether this choice changes if we have a collaborative third party, C , who reliably reports a measurement outcome on the third qubit.

1. Quantify the entanglement yield as the probability-weighted average of that present in each post-measurement pure state (pick your favorite $E(|\psi\rangle\langle\psi|)$ pure-state entanglement measure),

$$E_{\text{yield}}(\{p_i, |\psi_i\rangle\langle\psi_i|\}) = \sum_i p_i E(|\psi_i\rangle\langle\psi_i|) \quad . \quad (37)$$

What is the A - B entanglement yield for the GHZ and W states upon measurement of C in the Z -basis?

2. Tracing of a qubit may be implemented as the result-agnostic accumulation of states after measurement in any basis, i.e., the reduced density matrix is invariant with respect to the tracing basis. However, the measurement basis can affect the particular pure state decomposition remaining (providing physical intuition for the many-to-one relationship between pure state decompositions and pure states).

For both the $|GHZ\rangle$ and $|W\rangle$ state, find and describe a measurement basis for C that maximizes the entanglement yield established between A and B (note: your particular values

may depend on your chosen pure state entanglement measure). How do the maxima compare between the GHZ and W resource?

Hint: Intuitively, the property of being in the $|000\rangle$ or $|111\rangle$ state has been distributed throughout the three Hilbert spaces in the GHZ state. In efforts to *not* learn this information from the measurement of qubit C , recall from the previous exercise that minimally informative/uniform probability vectors are available upon a change of measurement basis.

3. Based on your experience, describe a C -collaborative protocol that begins in the $|W\rangle$ or $|GHZ\rangle$ state and deterministically produces the state $|\psi\rangle_{AB} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

Hint: You may wish to consider a two-step procedure of the form (1) C -space measurement and (2) measurement-dependent A, B -space local unitaries.

4. **Entangling from a Distance** You have two immobile qubits, labeled A_i and B_i , that you would like to entangle. Direct interactions between A_i and B_i are experimentally inaccessible. However, each immobile qubit can be entangled with a mobile qubit. These mobile qubits, A_m and B_m , may be brought together for direct interaction.

Devise a protocol within these constraints for the production of the state $|\psi\rangle_{A_i B_i} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

D. mixed-state entanglement distribution

The text remarks “[T]wo mixtures with the same density matrices are indistinguishable or equivalent (analogous to the way two pure states that differ only by a global phase are equivalent)”. In this problem, we will scrutinize to what extent this analogy holds in the context of mixed state entanglement structure.

1. Discuss the physical meaning captured by the irrelevance of global phases.
2. Consider this separable mixture of unentangled states (*Phys. Rev. Lett.* **91**, 037902),

$$\rho_{abc} = \frac{1}{6} \sum_{k=0}^3 |\Psi_k\rangle\langle\Psi_k| \otimes |\Psi_{-k}\rangle\langle\Psi_{-k}| \otimes |0\rangle\langle 0| + \frac{1}{6} \sum_{j=0}^1 |j\rangle\langle j| \otimes |j\rangle\langle j| \otimes |1\rangle\langle 1| \quad , \quad (38)$$

where the eigenvectors of the X, Y Pauli operators are parameterized as

$$|\Psi_k\rangle = \frac{|0\rangle + e^{ik\pi/2}|1\rangle}{\sqrt{2}} \quad . \quad (39)$$

List the six pure states present in the noted pure state decomposition.

3. Label Hilbert spaces from left-to-right as A, B , and C . Consider a procedure for distributing entanglement between parties A and B through direct interactions only with system C by sequential application of the entangling operation $CNOT_{control, target}$: first $CNOT_{A,C}$ followed by $CNOT_{B,C}$. You may imagine a scenario of entangling two stationary Hilbert spaces, A and B , through local interactions with a mobile Hilbert space, C , providing the interconnect.
 - Through application of $CNOT_{A,C}$ on each pure state in the initial decomposition, show that the state after the first entangling operation is

$$\rho'_{ABC} = \frac{1}{6} \sum_{k=0}^3 \rho_{AC} \left(\frac{|00\rangle + e^{ik\pi/2}|11\rangle}{\sqrt{2}} \right) \otimes \rho_B(|\Psi_{-k}\rangle) + \frac{1}{6} |0, 0, 1\rangle\langle 0, 0, 1| + \frac{1}{6} |1, 1, 0\rangle\langle 1, 1, 0| \quad , \quad (40)$$

where $\rho(|\psi\rangle) = |\psi\rangle\langle\psi|$. Are there entangled pure states present in the resulting pure state convex decomposition?

The terminology of mixed state entanglement is the following: If a convex decomposition (positive real weights $\sum_i p_i = 1$) of the density matrix composed of only tensor-product pure states exists,

$$\rho_{\text{separable}} = \sum_i p_i \rho_i^A \otimes \rho_i^B \quad , \quad (41)$$

then the mixed state is considered separable i.e., its density matrix could be *independently* created without distributing entanglement. A mixed state is said to be entangled *iff* it is not separable.

Is the resulting state, ρ'_{ABC} , entangled along the $AC|B$ Hilbert space partition? By analyzing the symmetry of the density matrix under the interchange of Hilbert spaces B and C , is the state considered entangled along the $AB|C$ Hilbert space partition?

- After application of the second entangling operation $CNOT_{B,C}$, show that the final mixed state is

$$\begin{aligned} \rho''_{ABC} = \frac{1}{6} \sum_{k=0}^3 \rho \left(\frac{1}{\sqrt{2}} \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \right) |0\rangle_C + \frac{1}{\sqrt{2}} \left(\frac{e^{-ik\pi/2}|01\rangle + e^{ik\pi/2}|10\rangle}{\sqrt{2}} \right) |1\rangle_C \right) \\ + \frac{1}{6} \rho(|0, 0, 1\rangle) + \frac{1}{6} \rho(|1, 1, 1\rangle) \quad . \quad (42) \end{aligned}$$

Using computational basis state projectors in the C Hilbert space, show that this density matrix is equivalent to one decomposed in the form

$$\rho''_{ABC} = p_0 \rho_{AB}^{(0)} \otimes |0\rangle\langle 0| + p_1 \rho_{AB}^{(1)} \otimes |1\rangle\langle 1| \quad , \quad (43)$$

and is thus considered separable along the $AB|C$ Hilbert space partition. Write the expressions for $\rho_{AB}^{(0)}$ and $\rho_{AB}^{(1)}$, are either of these entangled mixed states? Write the final pure state decompositions upon projective measurement of $|0\rangle_C$ or $|1\rangle_C$, with what probabilities would these two measurements be made? are entangled states present across the $A|B$ partition in either measurement outcome?

4. With the understanding developed above, contextualize the conclusion of *Phys. Rev. Lett.* **91**, 037902 that “no entanglement is necessary to distribute entanglement”. To what extent does the analogy hold between 1.) the non-uniqueness of pure-state decompositions for mixed states and 2.) the equivalence of pure states differing by a global phase. Would insertions of a global phase or exchanges of convex decompositions at intermediate points in a QI protocol be observable?

VI. GLOSSARY

In significant detail, *physically* and *mathematically* describe:

1. the role, importance, and properties of
 - wavefunctions (continuous and discrete)
 - superpositions

- entanglement
- density matrices

Explain why the colloquial definition of superposition as “a phenomenon in which a particle can be in two states at the same time” lacks mathematical clarity.

2. the distinction between pure and mixed quantum states
 - Describe the role of mixed states in incorporating noise or environmental interactions into simulations of quantum systems.
 - If quantum systems could be perfectly isolated from their environments and manipulated with exactly unitary operators, describe any residual value provided by the mixed state formalism. What kind of states do local operators in entangled states see?
3. the process of quantum measurement (projective, complete/partial, destructive/non-destructive)
 - Describe an application in which a partial projective measurement may be computationally valuable for quantum information.
 - [*KLM Exercise 3.4.4*] Explain why performing a complete Von Neumann measurement with respect to the computational basis and subsequently computing the parity of the resulting string is not equivalent to performing a projective measurement of the parity.